

RESEARCH ARTICLE

Automated detection of cephalometric landmarks using deep neural patchworks

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Objectives This study evaluated the accuracy of deep neural patchworks (DNPs), a deep learning-based segmentation framework, for automated identification of 60 cephalometric landmarks (bone-, soft tissue- and tooth-landmarks) on CT scans. The aim was to determine whether DNP could be used for routine three-dimensional cephalometric analysis in diagnostics and treatment planning in orthognathic surgery and orthodontics.

Methods: Full skull CT scans of 30 adult patients (18 female, 12 male, mean age 35.6 years) were randomly divided into a training and test data set (each $n = 15$). Clinician A annotated 60 landmarks in all 30 CT scans. Clinician B annotated 60 landmarks in the test data set only. The DNP was trained using spherical segmentations of the adjacent tissue for each landmark. Automated landmark predictions in the separate test data set were created by calculating the center of mass of the predictions. The accuracy of the method was evaluated by comparing these annotations to the manual annotations.

Results: The DNP was successfully trained to identify all 60 landmarks. The mean error of our method was 1.94 mm (SD 1.45 mm) compared to a mean error of 1.32 mm (SD 1.08 mm) for manual annotations. The minimum error was found for landmarks ANS 1.11 mm, SN 1.2 mm, and CP_R 1.25 mm.

Conclusion: The DNP-algorithm was able to accurately identify cephalometric landmarks with mean errors <2 mm. This method could improve the workflow of cephalometric analysis in orthodontics and orthognathic surgery. Low training requirements while still accomplishing high precision make this method particularly promising for clinical use.

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Objectives

Cephalometric analysis is commonly used in diagnosis and treatment planning of orthodontics and

orthognathic surgery cases. In the past, two-dimensional (2D) lateral radiographs were used for manual cephalometric analysis.¹ However, evolving three-dimensional (3D) imaging techniques have led to the development of 3D cephalometric analysis.² Accurate 3D analysis plays a crucial role in diagnosing craniofacial deformities,

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assessing their severity and virtual planning of different treatment options.^{3,4} Tondin *et al* found that the differences between pre-operative planning and post-operative outcomes of orthognathic surgeries were much lower for virtual planning compared to classical, analog planning.⁵ The quality of the 3D cephalometric analysis and consequently of the virtual planning process depends on the accuracy and reproducibility of landmark identification.⁶ Manual landmark identification requires medical expertise and training to achieve reliable results.² Moreover, manual placement of landmarks is prone to operator variability, leading to limited inter- and intraexaminer reliability and repeatability.^{3,7,8} The manual landmarking process is also time-consuming, with the minimum amount of crucial landmarks requiring at least 15 min to identify.^{2,9} As a result, cephalometric analysis is often not used in orthodontic and orthognathic planning.^{6,7,10–12} A fast and viable method for automated landmark identification is needed to save time and allow untrained clinicians to routinely use 3D cephalometric analysis.⁶

A systematic review by Dot *et al* evaluated the accuracy and reliability of various methods used for automated cephalometric landmark identification.² The review found that knowledge- and atlas-based methods have certain advantages, such as requiring little *a priori* information, being customizable, and being less complex.² However, atlas-based methods are sensitive to biological variations (*e.g.* sex, age) and their performance is highly dependent on registration techniques. Knowledge-based methods may perform poorly on landmarks located on curved structures.² In contrast, learning-based methods, such as fully convolutional neural networks (FCNs), have shown higher robustness and lower sensitivity to artifacts and noise in images.¹³ FCN-based methods allow for the representation of the heterogeneity of the appearance of each landmark.¹³ Entire volumes can be efficiently analyzed to detect and locate multiple landmarks simultaneously using a learned, common feature representation.¹³ Learning-based methods are especially suitable for the craniomaxillofacial region due to the large anatomical variability of cranial bones. Literature has shown promising results for automated landmark detection using learning-based methods, although many studies require hundreds of training data sets to identify only a few landmarks in specific regions.^{9,14} Additionally, most methods only include the automated identification of bone landmarks, with only two studies including dental landmarks.⁹ The aim of this study is to develop a robust and reliable method that can automatically identify a large number of landmarks. By introducing a novel method which only requires a few cases as a training data set, automated landmark identification can easily be adapted to include or change landmarks as needed, depending on the region of interest. By eliminating human subjectivity in the manual annotation process, better reliability and reproducibility of landmark identification can be

achieved, improving diagnosis and treatment planning based on 3D cephalometry.^{6,8}

Methods

This study was conducted in accordance with the Helsinki Declaration and received ethical approval from the Ethics Committee of the University of Freiburg (No.21–1293). All patients provided written informed consent for their data to be used in this study. The study included anonymized CT scans of 30 adult patients (18 female, 12 male, mean age 35.6 years, standard deviation = 15.3 years). The CT scans were obtained between March and September 2020 at the University Hospital in Freiburg i. Br. on a Siemens Somatom Definition Flash CT system, and covered the entire skull down to the lowest point of the mandible. Patients were asked to bring their teeth in maximum intercuspation. As scans were acquired with different settings for the head and midface, they were stitched together. Exclusion criteria were traumatic fractures, severe skeletal anomalies, tumor diseases, extensive artifacts caused by dental metal restorations, and edentulous patients.

Based on studies by Zamora *et al*, Santos *et al* and Codari *et al*, 60 landmarks were chosen with regard to their clinical importance.^{15–17} The landmarks were divided into four groups based on their anatomical location: bone landmarks (Group 1, $n = 22$), soft tissue landmarks (Group 2, $n = 18$), maxillary dental landmarks (Group 3, $n = 10$), and mandibular dental landmarks (Group 4, $n = 10$). Table 1 (Appendix) lists all landmarks and their corresponding anatomical locations.^{15–17} The study included a training data set of 15 cases and a separate test data set of 15 cases. Two clinicians (A and B) were chosen to annotate all landmarks. Both clinicians were maxillofacial residents with equal experience in identifying landmarks. Manual landmark identification was performed using the open source software 3D Slicer (v. 4.10.2) on multiplanar reconstructions (MRPs), with cross-verification on the volume-rendered 3D image to increase precision.^{18,19} The selected cephalometric landmarks were manually traced by Clinician A in all 30 CT scans and were considered as ground truth. Clinician B annotated the test data set of 15 CTs only. Afterwards, the data were imported onto the local medical imaging platform NORA (University of Freiburg - Nora Medical Imaging Platform Project 2022) for training of the DNP.

Deep learning framework and data pipeline

The in-house developed deep learning framework and data pipeline used in this study is deep neural patchworks (DNPs). In previous studies conducted by our research team, DNP has been demonstrated to be effective for automated segmentation of head CT scans and enables the use of 3D segmentation in high-resolution CTs regardless of input size.^{20,21}

Table 1 Cephalometric landmarks with descriptions of their positions

<i>Landmark</i>	<i>Position</i>
STN	Soft tissue nasion
IC_R	Inner canthus right
OC_R	Outer canthus right
IC_L	Inner canthus left
OC_L	Outer canthus left
NT	Tip of nose
SN	Subnasale
STA	Soft tissue A-point
Ala_R	Ala: the extreme medial point of ala right
Ala_L	Ala: the extreme medial point of ala left
MA_R	Right commissure
MA_L	Left commissure
UL	Upper lip: medial point of vermillion border
LL	Lower lip: medial point of vermillion border
STB	Soft tissue B-point
STPg	Soft tissue pogonion
T_R	Tragus (the extreme posterior point) right
T_L	Tragus (the extreme posterior point) left
N	Nasion: the extreme anterior point on the fronto-nasal suture
Fz_R	Sutura frontozygomatica: the extreme anterior, inferior, medial point of the sutura frontozygomatica right
Fz_L	Sutura frontozygomatica: the extreme anterior, inferior, medial point of the sutura frontozygomatica left
Or_R	Orbitale: the deepest point on the infraorbital margin right
Or_L	Orbitale: the deepest point on the infraorbital margin left
ANS	Anterior nasal spine: the extreme anterior point on the maxilla
PNS	Posterior nasal spine: the extreme posterior point on the maxilla
A	Point A: the deepest point on the curvature of the maxillary alveolar process
B	Point B: the deepest point on the curvature of the mandibular alveolar process
Pg	Pogonion: the extreme anterior point of the mandibular symphysis
Por_R	Porion: the extreme superior point of meatus acusticus externus right
Co_R	Condylion: the extreme superior point on the condylar head right
CoLat_R	Condylion lat.: the extreme lateral point on the condylar head right
CoMed_R	Condylion med.: the extreme medial point on the condylar head right
CP_R	Proc. coronoideus: the extreme superior, anterior point on the processus coronoideus right
Por_L	Porion: the extreme superior point of meatus acusticus externus left
Co_L	Condylion: the extreme superior point on the condylar head left
CoLat_L	Condylion lat.: the extreme lateral point on the condylar head left
CoMed_L	Condylion med.: the extreme medial point on the condylar head left
CP_L	Proc. coronoideus: the extreme superior, anterior point on the processus coronoideus left
S	Sella: the midpoint of sella turcica
Ba	Basion: the most inferior posterior point on the sagittal plane on the anterior border of the foramen magnum
13A	Center of apex of 13
11A	Center of apex of 11
21A	Center of apex of 21
23A	Center of apex of 23
16C	Mesiobuccal cusp tip of 16
13C	Canine cusp tip of 13
11C	Extreme superior, medial point of crown of 11
21C	Extreme superior, medial point of crown of 21
23C	Canine cusp tip of 23
26C	Mesiobuccal cusp tip of 26
46C	Mesiobuccal cusp tip of 46

(Continued)

Table 1 (Continued)

Landmark	Position
43C	Canine cusp tip of 43
41C	Extreme superior, medial point of crown of 41
31C	Extreme superior, medial point of crown of 31
33C	Canine cusp tip of 33
36C	Mesiobuccal cusp tip of 36
43A	Center of apex of 43
41A	Center of apex of 41
31A	Center of apex of 31
33A	Center of apex of 33

To prepare the training data set for a segmentation task, spheres with a radius of 2mm are constructed around each landmark and multiple stacks of nested patches with fixed matrix size 32^3 are drawn as training batches in a scale pyramid of depth four. The physical patch size of the coarsest scale is 100mm, while the finest has an isotropic volume of 32mm (and therefore 1mm^3 isotropic resolution), with intermediate levels exponentially interpolated. The input values for the CT images are based on Hounsfield units (HUs), which are separated into 11 feature channels using an adaptable channel separation layer, allowing automated segmentation in different grayscale ranges without the need of manual definition. These data are used to train the network on detecting landmarks through a segmentation-based approach using a stack of 3D convolutional neural networks (CNNs) developed on the basis of the U-Net architecture.²²

At each batch scale, the U-Net-like architecture is trained, with input matrix sizes of isotropic patches and feature dimensions of [8,16,16,32,64]. Max-pooling is used in the encoding layers and transposed convolutions in the decoding layers, while connections between different scales are based on $n + 8$ output channels, where n corresponds to the number of landmarks. These connections are a defining feature of the architecture, enabling intermediate loss calculations. The networks are using the Adam optimizer with a learning rate of 0.001.²³ The loss function is a categorical cross-entropy using a candidate sampling approach.²⁴ No specific hyperparameter tuning is performed, and existing values based on prior applications of generic medical segmentation tasks are used.²⁰ The integrated augmentation method includes a random probability of rotation up to 20° and deformation up to 20% in each axis while drawing patches. Each network is trained for 5 million training steps in approximately 24h using a GPU (RTX A6000 48 GB Nvidia, Santa Clara, CA) accelerated server for each of the four landmark tasks. The output of each network is a segmentation of the region based on the sphere around the landmark, with the predicted landmark calculated as the center of mass of the middle of the segmented region.

Metrics

To assess the performance of the AI method, the Euclidean distance between the AI-generated landmarks and manual landmarks considered as ground truth was calculated. The mean, median, and standard deviation of these values were then analyzed to determine the overall accuracy of the method and additionally success detection rates (SDRs) using precision ranges of $<2\text{mm}$ and $<4\text{mm}$, were calculated. The Euclidean distance between the ground truth and the landmarks placed by Clinician B was also evaluated and mean, median, and standard deviations were calculated. Boxplots were created for each class to further evaluate the results.

Code availability

The framework for recreating the presented landmark detection model is available on an open Git repository (Bitbucket: <https://bitbucket.org/reisert/patchwork>). Further information regarding the segmentation approach is presented as wiki (<https://bitbucket.org/reisert/patchwork/wiki/Home>) and as preprint. A technical description is available on Arxiv (<https://arxiv.org/abs/2206.03210>). Further code used in our analysis is available from the corresponding author upon reasonable request.

Results

Training and testing

In this study, a fully automated AI-based 3D segmentation algorithm was developed for focal segmentation using the DNP method to identify 60 different cephalometric landmarks. The algorithm was able to localize all 900 landmarks in the separate test cohort of 15 patients using a patch-based prediction method with secondary calculation of the mean position within the predicted segmentations for each landmark. Automated landmark identification using the developed algorithm took approximately 2 min per CT scan, depending on the image volume size, without using graphic unit acceleration.

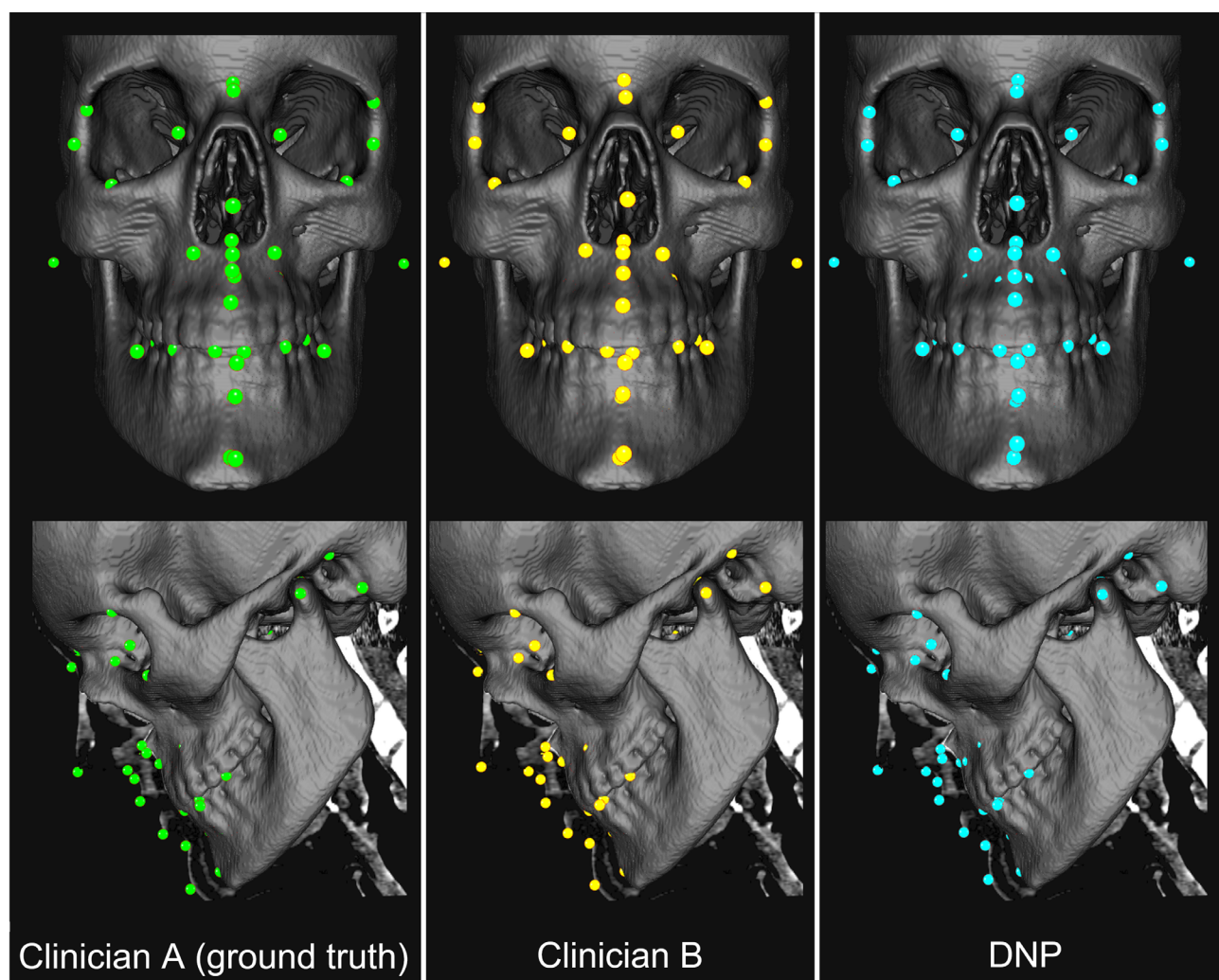


Figure 1 Manually annotated ground truth landmarks (green), landmarks annotated by Clinician B (yellow) and predicted landmarks by the DNP algorithm (blue). CT segmentations were automatically performed by the proposed deep learning method (NORA, University of Freiburg).

Accuracy of landmark identification

The proposed method for automated landmark identification achieved a mean localization error of 1.94 mm (SD, 1.45 mm). This error was calculated as the mean distance between the automatically identified landmark and the ground truth set by Clinician A (Figure 1). The mean localization error for the manual annotation of Clinician B was 1.32 mm (SD 1.08 mm), which is comparable to automated landmark localization. The SDRs of our automated method were 66.4 and 91.9% using precision ranges of <2 mm and <4 mm, respectively. Figure 2 and Figure 3 in the Appendix shows the mean errors, minimum and maximum errors, and SDRs for both the automated and manual identification of landmarks for all 60 landmarks.

As shown in Figure 4, the mean errors for the automatically identified bone landmarks (mean error

1.83 mm, SD 1.29 mm) were lower than those for the soft tissue landmarks (mean error 1.91 mm, SD 1.6 mm) and dental landmarks (mean error 2.08 mm, SD 1.45 mm). When further dividing dental landmarks into maxillary dental landmarks (mean error 1.98 mm, SD 1.42 mm) and mandibular dental landmarks (mean error 2.18 mm, SD 1.47 mm), the latter showed the highest mean error of all landmark groups. In comparison, the manually identified landmarks had the lowest mean errors in the dental landmarks (mean error 0.95 mm, SD 0.65 mm), with mean errors of 0.85 mm (SD 0.59 mm) for the maxillary dental landmarks and 1.04 mm (SD 0.69 mm) for the mandibular dental landmarks. The mean errors for the soft tissue landmarks (mean error 1.38 mm, SD 1.08 mm) and bone landmarks (mean error 1.63 mm, SD 1.28 mm) were also lower than those for the automatically identified landmarks.

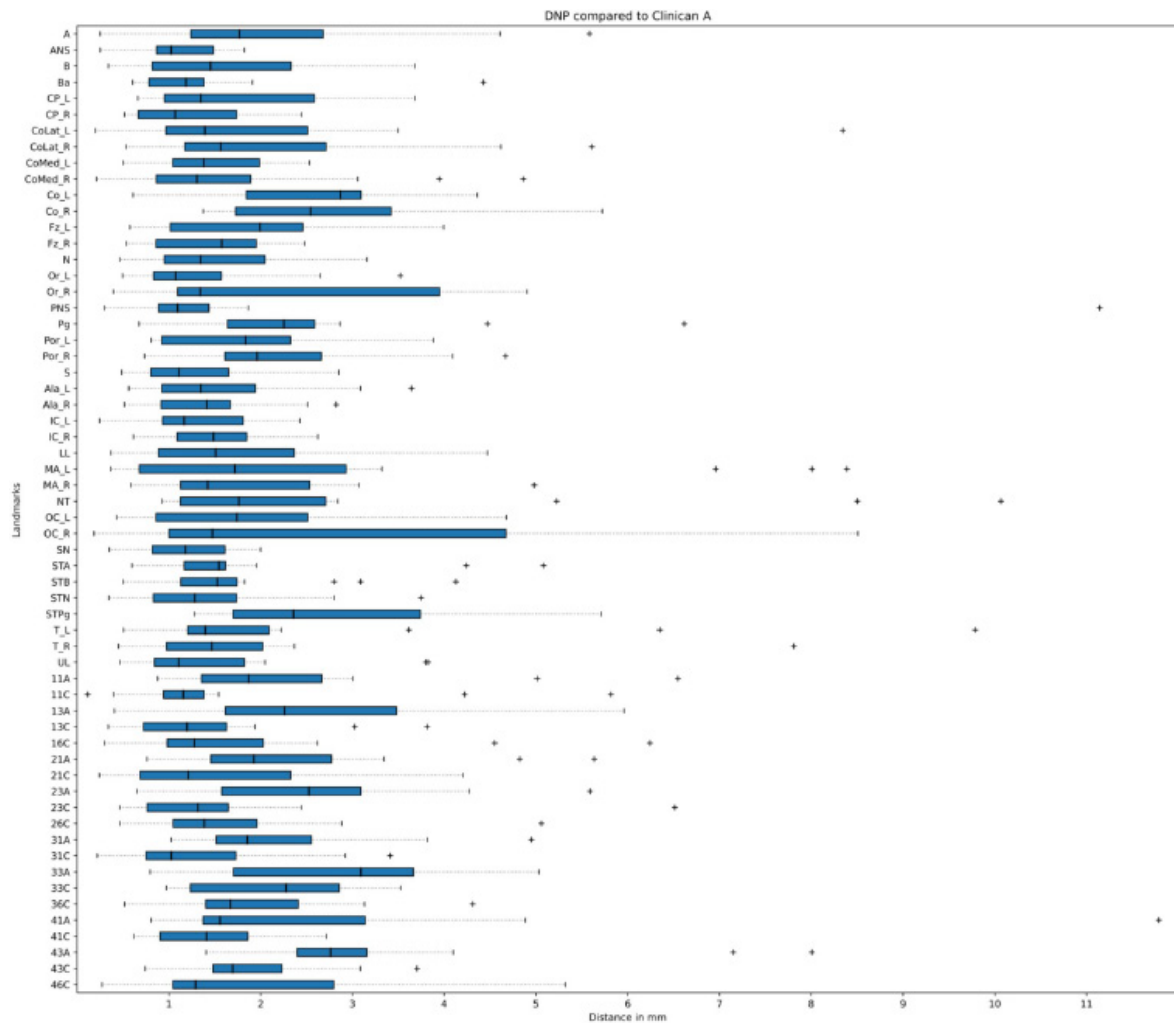


Figure 2 Mean errors, minimum errors and maximum errors of all landmarks annotated by the DNP algorithm compared to the manual annotations of Clinician A (NORA, University of Freiburg).

Minimum and maximum errors

When considering each landmark individually, we found that the greatest discrepancy in the automatically identified bone landmarks was for landmark Co_R (the extreme bone superior point on the condylar head right), with a mean error of 2.77 mm (SD 1.25 mm). The greatest error in the soft tissue landmarks was found for soft tissue pogonion (STPg), with a mean error of 2.93 mm (SD 1.38 mm). The center of apex of 43 (43A) showed the highest error in the dental landmarks (mean error 3.26 mm, SD 1.84 mm).

On the other hand, the best agreement in localization among the bone landmarks was found for the anterior nasal spine (ANS) (mean error 1.11 mm, SD 0.46 mm), among the soft tissue landmarks for subnasale (SN) (mean error 1.2 mm, SD 0.49 mm), and among the dental landmarks for the extreme superior, medial point of crown of 31 (31C) (mean error 1.37 mm, SD 0.89 mm). All minimum and maximum errors can be found in Figure 4 Appendix.

Discussion

The development of automated landmarking methods is a growing area of research that has the potential to improve the efficiency and accuracy of cephalometric analysis. Fully automated 3D landmark annotation is a relatively new technology, but it has already demonstrated promising results in terms of saving computational time and improving reproducibility for various orthodontic and maxillofacial tasks.³

Studies have shown that facilitating cephalometric analysis, which is currently very time-consuming, leads to broader use in clinical practice, improving the accuracy of diagnosis, pre-operative planning, and outcomes.³ In particular, our novel automated landmarking approach could improve the accuracy of skeletal and facial type definitions, symmetry evaluations, and evaluation of maxillary-mandibular relationships. Although there are many techniques available to make

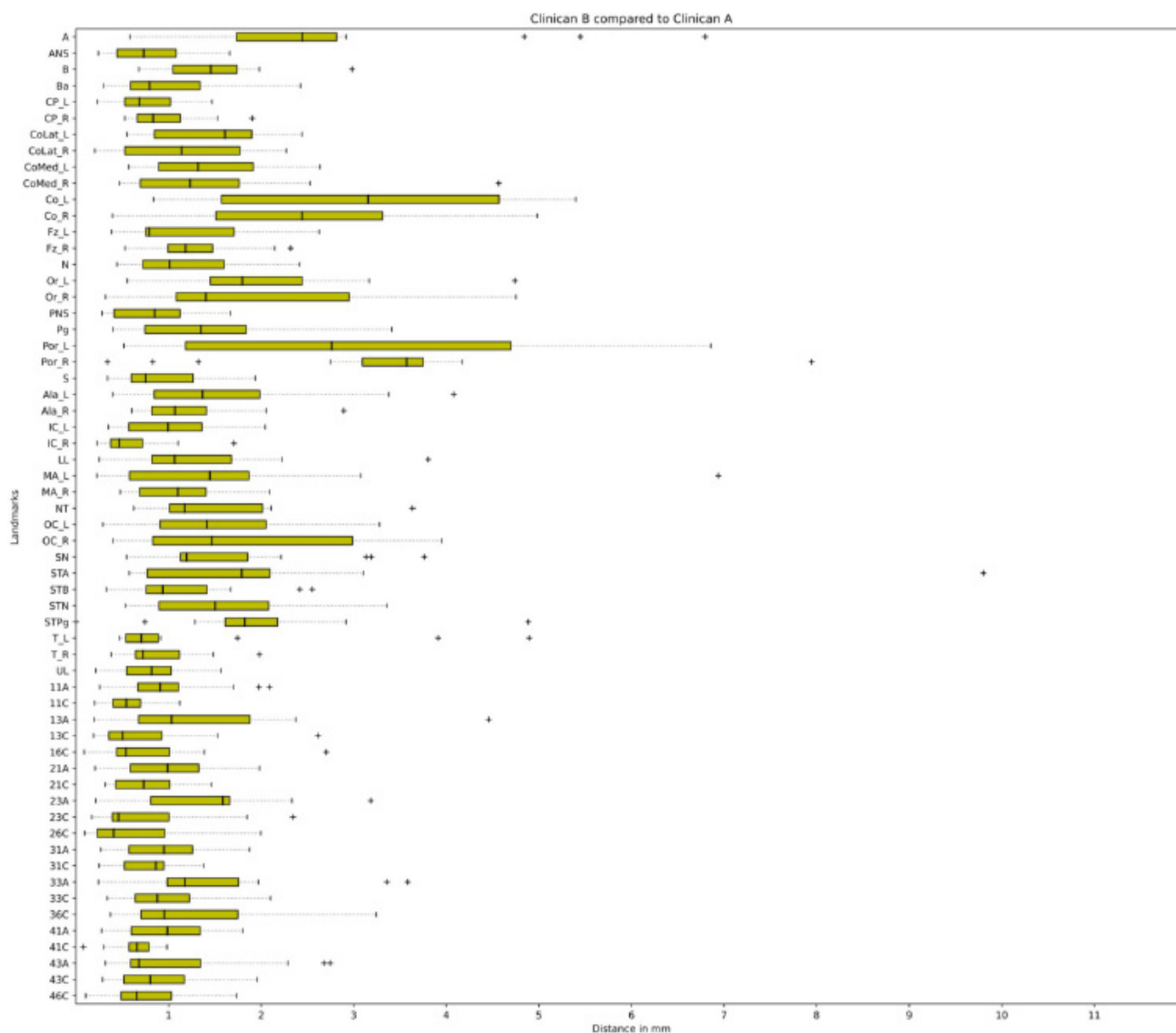


Figure 3 Mean errors, minimum errors and maximum errors of all landmarks manually annotated by Clinician B compared to the manual annotations of Clinician A (NORA, University of Freiburg).

the process of cephalometric analysis easier, most of them require either hundreds of training data sets, manual placement of at least a few landmarks, or have insufficient performance in placing landmarks, leading to inaccurate results.⁶ Fully automated 3D landmark annotation remains a challenging problem in the field.

This study presents a novel approach, DNPs, for automated landmark detection based on a region of interest segmentation.²⁰ Using the proposed method, all 900 landmarks (60 landmarks per patient in 15 patients) were successfully identified. To evaluate the accuracy of the automated landmarks, we compared their distances to manually annotated landmarks. In this study, the mean error was less than 2 mm for all 60 landmarks, demonstrating the high accuracy of our method and its potential for clinical applications. Previous research has

indicated that an error of 4 mm is acceptable for clinical diagnosis on 2D X-ray images.^{10,25} While the limits for 3D images have yet to be defined, multiple studies have considered mean errors of 2 mm as clinically acceptable.^{10,25–27}

The proposed 3D CNN architecture is hierarchical and patch-based, using a stack of CNNs based on the U-Net architecture. One key characteristic of the proposed 3D CNN architecture is that it processes image data patchwise in connected levels of image resolution, which allows for the consideration of varying coverage in medical imaging based on scan length and field of view (FOV). This approach aims to reduce the computational requirements for 3D CNNs while maintaining high resolution of the scans. Therefore, memory limitations of GPUs that are one of the main limitation

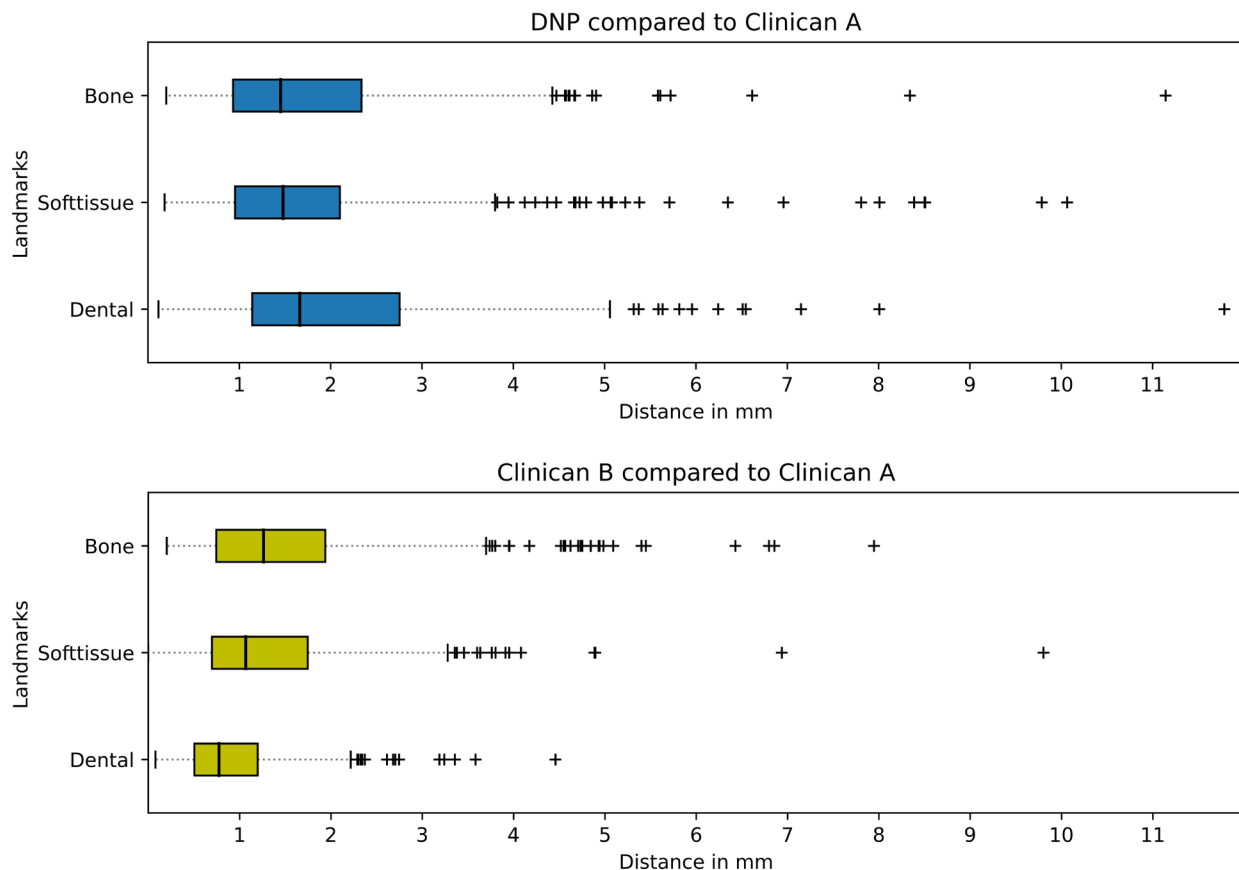


Figure 4 Mean localization errors (mm) of bone landmarks, soft tissue landmarks and dental landmarks for automatically identified landmarks (blue) and manually identified landmarks (green) (NORA, University of Freiburg).

in this field of research can be avoided without limiting the high-resolution outputs. Other approaches usually address this issue by downsampling the images or reducing their network input resolution in segmentation tasks, which can result in less accurate landmark identification.²⁸ Alternatively, CNNs were implemented on small 3D image patches, which preserves the resolution of the scan, however lacks volumetric context.²⁹ Using this combination of input both high resolution features and coarse relationships are represented in the trained networks, such as the position of the pogonion landmark relative to the menton landmark, while still accounting for the geometric variability of edge structures. Furthermore, this novel method used balancing and augmentation techniques during patch-based sampling of the data to accurately represent small landmarks in the training task, with none of the labels missed in the prediction.

The results of this study favorably compare to both CNN and non-CNN approaches, despite using fewer training data sets and a larger number of landmarks. A systematic review by Dot et al. 2020 found that deep learning-based methods for automated landmarking

proved to be the most accurate, with multiple studies reporting mean errors <2 mm.² However, these studies focused on only a few landmarks. For example, De Jong et al reported that 9 out of 33 landmarks had a mean error <2 mm, while Torosdagli et al reported mean errors of <0.5 mm, but in their study, only eight landmarks were placed automatically.³ In a very recent study conducted by Ghowasi et al, 53 landmarks, including bony landmarks as well as dental landmarks were automatically annotated.³⁰ However, their reported mean error distance for all 53 landmarks was 3.19 mm (SD, 2.6 mm), which lies well above the mentioned mean error of 2 mm stated to be clinically acceptable.³⁰

None of the existing studies automatically localized a total of 60 landmarks, all of which showed mean errors <2 mm. Considering the high anatomical variability in facial bones, relying on a small number of landmarks will inevitably lead to inaccuracy in analysis, consequent diagnosis and treatment planning, and outcomes. Therefore, a broad selection of both mid-sagittal landmarks as well as bilateral landmarks is needed to properly display the anatomical shape of the cranial bones.

In this study, especially good results were found for landmarks based on bony structures. This has also been reported in literature. Gupta *et al* stated a minimum mean error of 1.17 mm (SD, 0.49 mm) for the landmark “nasion”, and Shahidi *et al* reported a minimum mean error of 0.89 mm (SD, 0.62 mm) for the landmark “A-point”.¹⁰ Minimum mean errors in this study were found at the landmarks “ANS”, “SN”, and “CP_R” with mean errors of 1.11 mm (SD, 0.46 mm), 1.2 mm (SD, 0.49 mm) and 1.25 mm (SD, 0.6 mm).

Larger errors were found for dental landmarks as these landmarks are often closely spaced and for soft tissue landmarks as they are often located on curved structures [e.g. *soft tissue pogonion (STPg)*]. The greatest errors were found for the landmarks “43A”, “STPg”, and “NT” with errors of 2.85–3.26 mm. Only few studies include soft tissue and dental landmarks as they focus on bony landmarks used for basic cephalometric analyses. To our knowledge, there are only two studies on automated 3D landmarking that include multiple dental landmarks so far.^{26,29} However, especially regarding aesthetic aspects, soft tissue landmarks play an important role in both diagnosis and treatment planning. As the position of the teeth greatly influence the course of treatment, they should also be included in 3D analysis. Therefore, even though soft tissue landmarks and dental landmarks are more error prone in automated landmarking, we chose to not only include bony landmarks but also soft tissue and dental landmarks in this study.

It is worth noting that our method was only trained on a total of 15 image data sets, which is a significantly lower training burden compared to other studies in the field.^{2,3,7,12,13} In comparison, Ghowsi *et al* used 100 CBCT scans to train their automated landmark identification system resulting in mean errors of 3.19 mm (SD, 2.6 mm). The exceptionally low training burden is a great advantage of this method especially because collection of data in the medical and dental field comes with special hurdles.³¹ Medical records or images are often not easily accessible due to strict patient data policies, lacking patient consent or non-compatible imaging systems.^{31,32} Compared to other data sets used in the AI realm, e.g. advertisement, search engines, route planners, etc. datasets in the medical field are often very small.^{31,32}

Additionally, the proposed method is generic and can be easily adapted to include or change landmarks as needed, depending on the region of interest or the type of study.¹² For instance, the landmarks used in this study were set and tested in a patient population consisting of a wide range of ages across both sexes, including a range of anatomical and morphological variations as seen in Freiburg, Germany. As a result, our database mainly consists of western European individuals, and the results may not be applicable to other populations. Furthermore, the exclusion criteria applied in this study resulted in a training data set without any craniomaxillofacial deformities such as fractures or underlying

diseases, missing bones or teeth, or large artifacts caused by dental metal restorations, implants, or surgical plates and screws.^{3,4} Since cephalometric analysis is often used in adolescent patients, the initial aim was the automated identification of landmarks in a healthy patient population. However, given the low training burden of the DNP only a few CTs of other ethnical groups, different landmarks or patients with pathologies are required to retrain the DNP accordingly. The proposed method can also be modified to identify landmarks in other imaging modalities, such as cone beam computed tomography (CBCT). The CT scans used in this study were obtained for routine or emergency diagnostic purposes, but not specifically for 3D cephalometric analysis. CT scans involve high radiation exposures and are therefore not a suitable imaging technique for cephalometric analysis.⁸ CBCTs, on the other hand, involve lower levels of radiation exposure and may be justified in cases requiring a combination of orthodontic and surgical treatment.^{8,10,33,34} As stated above, only a few CBCTs are required to retrain the DNP to detect the given landmarks in CBCTs, reducing the radiation exposure for patients.

In this study, the annotations placed by Clinician A were taken as the ground truth for each landmark. Other studies have used the mean position of multiple manual annotations as their ground truth.²⁶ However, both approaches have the potential for subjectivity, as the definition of the ground truth is dependent on the human eye.¹⁰ Williams *et al* classified landmarks based on their characteristics of identification.^{10,35} Biological landmarks, such as the nasion, can be clearly identified, while constructed landmarks, such as the gonion, are more difficult to identify. Fuzzy landmarks, such as the pogonion, are most prone to error, as the precise location of a point on a curve (e.g. the highest point) is subjective to the clinician and dependent on the patient's head positioning.^{6,35} Given the variability in the identification of fuzzy and constructed landmarks among clinicians, neither the mean annotation of multiple clinicians nor the annotation of a single clinician can be considered the anatomically exact definition of landmarks. Given a questionable ground truth, large mean errors in automated landmarking may be caused by intra- and interrater disagreements rather than poor performance of the AI method. De Jong *et al* found that interrater performance was particularly poor for landmarks with difficult-to-reproduce anatomical definitions, and that the automated landmarking method had a clear advantage in reproducibly identifying these landmarks.^{10,12} However, mean errors calculated as deviations from a ground truth defined by the human eye are prone to subjectivity and error. Many cephalometric landmarks, such as the porion, are crucial for correct analysis but have poor reproducibility and tend to have high intra- and interrater variability and error.⁶ This highlights the importance of using objective and reliable methods to increase the accuracy and efficiency of automated

landmark identification and reduce human subjectivity in manual annotation.

Conclusion

In this study, we propose a new deep learning approach using DNPs, a hierarchical 3D patch-based segmentation framework, for large-scale landmark detection in cephalometric analysis. The advantage of this novel method is its ability to learn both high resolution features and coarse relationships while still accounting for the geometric variability of edge structures. This allows for more accurate and robust landmark detection

while reducing computational requirements at the same time. Trained on a small number of data sets, the DNP automatically identified 60 cephalometric landmarks with reliable results of mean errors <2mm, making it easier, faster, and potentially less prone to error than manual methods. The clinical application of this method could improve diagnostics and treatment planning in the craniomaxillofacial region, ultimately leading to improved patient outcomes. However, further research is needed to assess the potential for the generalization of these findings to different patient populations and to evaluate the performance of the method in severe pathological cases.

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